Karl Oskar Ekvall

Assistant Professor in Statistics
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Employment

University of Florida

Assistant Professor 2022–

Karolinska Institutet

Assistant Professor / Postdoctoral Researcher 2020–2022

TU Wien

Postdoctoral Researcher 2019–2020

Education

University of Minnesota – Twin Cities

Ph.D. in Statistics 2019

M.S. in Statistics 2017

University of Gothenburg

M.Sc. in Finance 2012

B.Sc. in Economics 2011

Research

My research focuses on reliable inference in nonstandard settings with, for example, complex dependence, parameters near the boundary, or singular Fisher information. I develop theory, methods, and computationally efficient software implementing the proposed methods. I regularly collaborate with researchers in biomedicine, psychology, and economics.

Publications

Published

Y. Zhang, K.O. Ekvall, and A.J. Molstad, 2026, "Universal inference for variance components". *Stat* 15(1): e70141

K.O. Ekvall and M. Bottai, 2026, "Uniform inference in linear mixed models". *Biometrika* 113(1): asaf079

Y. Zhang, K.O. Ekvall, and A.J. Molstad, 2025, "Fast and reliable confidence intervals for a variance component". *Biometrika* 112(2): asaf010

A.J. Molstad, K.O. Ekvall, and P.M. Suder, 2024, "Direct covariance matrix estimation with compositional data". *Electronic Journal of Statistics* 18(1): 1702–1748

K.O. Ekvall and M. Bottai, 2023, "Concave likelihood-based regression with finite-support response variables". *Biometrics* 79(3): 2286–2297

K. Gustin, K.O. Ekvall, M. Barman, B. Jacobsson, A. Sandin, A.-S. Sandberg, A.E. Wold, M. Vahter, and M. Kippler, 2023, "Mediation by thyroid hormone in the relationships between gestational exposure to methylmercury and birth size". *Exposure and Health* 16: 357–368

K.O. Ekvall and M. Bottai, 2022, "Confidence regions near singular information and boundary points with applications to mixed models". *Annals of Statistics* 50(3): 1806–1832

K.O. Ekvall, 2022, "Targeted principal components regression". *Journal of Multivariate Analysis* 190: 104995

K.O. Ekvall and A.J. Molstad, 2022, "Mixed-type multivariate response regression with covariance estimation". *Statistics in Medicine* 41(15): 2768–2785

K.O. Ekvall and G.L. Jones, 2021, "Convergence analysis of a collapsed Gibbs sampler for Bayesian vector autoregressions". *Electronic Journal of Statistics* 15(1): 691–721

K.O. Ekvall and G.L. Jones, 2020, "Consistent maximum likelihood estimation using subsets with applications to multivariate mixed models". *Annals of Statistics* 48(2): 932–952

Submitted

K.O. Ekvall, O.G.H. Hössjer, M. Bottai, and J.M.P. Albin. "Asymptotics for likelihood ratio tests of boundary points with singular information and unidentifiable nuisance parameters".

K.O. Ekvall. "Likelihood-based inference with separable correlation matrices".

C.N. Martinez-Perez, J. Williams, C.M. Ritchey, T. Kuroda, K.O. Ekvall, and C.A. Podlesnik. "Brief stimulus changes signaling extinction enhance resurgence".

H. Karim, E. Piscator, K.O. Ekvall, G. Riva, K.G. Lauridsen, and T. Djärv. "Validation and comparison of two prearrest prediction scores for in-hospital cardiac arrest survival".

Working papers (in preparation)

M. Shedden and K.O. Ekvall. "Score-based confidence intervals for variance-covariance parameters in linear mixed models".

M. Shedden and K.O. Ekvall. "Profile score asymptotics under near-boundary parameter sequences".

X. Wang, K.O. Ekvall, and A.J. Molstad. "Large covariance matrix estimation with spectral transfer learning".

Y. Kwon, K.O. Ekvall, B. Sherwood, and A.J. Molstad. "Fast algorithms for semi-supervised noncrossing multiple quantile regression in high dimensions".

K.L. Montague, C.N. Martinez-Perez, A. Edwards, C.M. Ritchey, M.S. Lamperski, T. Kuroda, K.O. Ekvall, and C.A. Podlesnik. "Evaluating multiple-context training to mitigate resurgence during context changes".

C.N. Martinez-Perez, K.L. Montague, C.M. Ritchey, A.G. Barnhart, T. Kuroda, S.P. Gilroy, K.O. Ekvall, and C.A. Podlesnik. "Evaluating control conditions when testing for resurgence".

D. Avila-Rozo, C.M. Ritchey, T. Kuroda, K.O. Ekvall, and C.A. Podlesnik. "Context change enhances resurgence with downshifts in alternative reinforcement".

A.M. Villalobos, D. Avila-Rozo, A. Molnar, G. Tirado, T. Kuroda, K.O. Ekvall, and C.A. Podlesnik. "Contingency discrimination: Evaluation of the effects of music and a contingency-discrimination criterion on resurgence".

Other scholarly contributions

K.O. Ekvall and G.L. Jones, 2019, "Markov chain Monte Carlo." *Wiley StatsRef*

Presentations

"Uniform inference near the boundary, with mixed models examples". 2025
EcoSta 2025, Waseda University, Tokyo, Japan

"Uniform inference near boundary and singular information points". 2025
Department of Mathematics, Stockholm University, Stockholm, Sweden

"Reliable inference in mixed models". 2025
Department of Epidemiology and Biostatistics, Karolinska Institutet, Stockholm, Sweden

"Confidence regions when the parameter is near the boundary". 2024
CMStatistics, London, U.K.

"Inference on some (nearly-)singular covariance matrices". 2022
CMStatistics, London, U.K.

"Inference on variance parameters near or at the boundary of the parameter set". 2022
University of Minnesota, School of Statistics anniversary. Minneapolis, MN, U.S.

"Reliable inference on small scale and variance parameters in mixed models". 2021
MEB biostatistics seminar. Stockholm, Sweden

"Confidence intervals for small scale parameters". 2020
IMM research day. Stockholm, Sweden

"Consistent maximum likelihood estimation in mixed models using subsets". 2020
Joint Statistical Meetings. Philadelphia, PA, U.S.

"Convergence analysis of a collapsed Gibbs sampler for Bayesian vector autoregressions". 2019
CMStatistics. London, U.K.

"Consistent maximum likelihood estimation in mixed models using subsets". 2019
University of Vienna seminar. Vienna, Austria

"Convergence analysis of a collapsed Gibbs sampler for Bayesian vector autoregressions". 2019
TU Wien colloquium. Vienna, Austria

"A multivariate linear model with separable correlation". 2017
International Chinese Statistical Association, applied statistics symposium. Chicago, IL, U.S.

Teaching

University of Florida

Applied Multivariate Statistics (undergraduate and graduate level) 2024–

Statistical Learning (undergraduate and graduate level) 2024–

Introduction to Probability (undergraduate level) 2022–2025

Introduction to Statistics Theory (undergraduate level) 2022–2024

Karolinska Institutet

Biostatistics (undergraduate level) 2020–2021

Interprofessional Learning Day (graduate and professional level) 2021

University of Minnesota – Twin Cities

Introduction for new teaching assistants (graduate level) 2018–2019

Theory of Statistics (undergraduate level) 2017–2018

Introduction to Statistical Computing (undergraduate level) 2018

Introduction to Statistical Analysis (as TA, undergraduate level) 2014–2016

Student Supervision

As advisor or co-advisor

Matias Shedden, Ph.D. in Statistics, University of Florida 2024–

Yiqiao Zhang, Ph.D. in Statistics, University of Florida (Now at Microsoft) 2022–2025

Jonatan Risberg, M.Sc. in Applied Mathematics, KTH (Summer research project at KI) 2021

As committee member

M.K. Kim (Ph.D. in Statistics, 2024–), K.M. Gelis Cadena (Ph.D. in Statistics, 2024–2025), X.H.F. Mak (Ph.D. in Statistics, 2025–), S. Li (Ph.D. in Food Science, 2024–2025)

Service

Editorial board

Associate Editor for Statistics and Probability Letters 2024–

Conference / session organization

Session EO195, The 9th International Conference on Econometrics and Statistics (EcoSta), Ryukoku University, Kyoto, Japan 2026

Reviewer

Annals of Statistics, Biometrika, Annals of Applied Statistics, Journal of the American Statistical Association, Statistica Sinica, Harvard Data Science Review, Journal of Internal Medicine, Statistical Methods in Medical Research, Statistics in Medicine, National Science Foundation, Electronic Journal of Statistics, Journal of Computational and Graphical Statistics, Computational Statistics and Data Analysis, Annales de l'Institut Henri Poincaré, Statistics

Internal

Faculty Search Committee 2024–2025

Department Executive Committee 2024–2025

Strategic Planning Committee 2025

Software

`reconf` R package for score-based confidence intervals for variance components in linear mixed models. https://github.com/koekvall/reconf `predglmm` R package for prediction in generalized linear mixed models. https://github.com/koekvall/predglmm

`sepcor` R package for estimating covariance matrices with separable correlation. https://github.com/koekvall/sepcor

`lmmstest` R package for a modified score test in linear mixed models. https://github.com/koekvall/lmmstest

`mmrr` R package for mixed-type multivariate response regressions. https://github.com/koekvall/mmrr

`fsnet` R package for elastic net-regularized regression with finite-support responses. https://github.com/koekvall/fsnet

`tpcr` R package for targeted principal components regressions. https://github.com/koekvall/tpcr

Honors and Awards

Graduate research partnership program fellowship	2017
The American–Scandinavian foundation fellowship	2016
Lynn Y.S. Lin fellowship for statistical consulting	2016
Fulbright foreign student program	2014
Tom Hedelius foundation scholarship	2014
Sixten Gemzéus foundation scholarship	2014
Malmsten award for best thesis in M.Sc. in Finance program	2014
School of Statistics first year scholarship	2014